

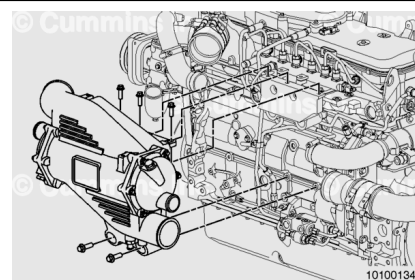
Trainings ▾

General Information

There are two aftercooler designs available based on the engine option.

Horizontal Aftercooler Assembly

The horizontal aftercooler assembly is a horizontal setup with the water inlet and outlet connections on the front and back. The horizontal aftercooler assembly is part of Cooling System Option 1.



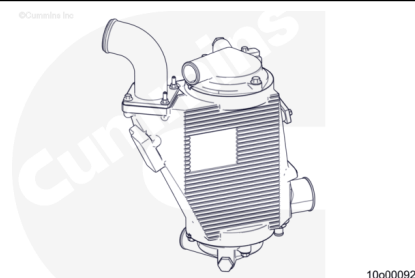
The aftercooler water inlet and outlet connections are made of plastic (1) or bronze (2) material.

The bronze and plastic water inlet and outlet can be identified by the location of the anode relative to the mounting capscrews. The plastic water inlet and outlet connections (1) have the anode on top of the mounting cap. The bronze water inlet and outlet connections (2) have the anode between the mounting capscrews.



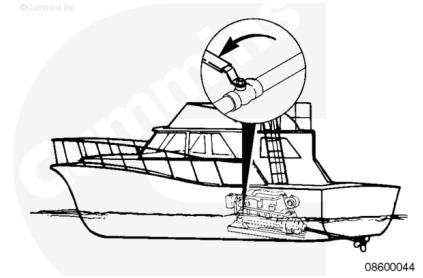
Vertical Aftercooler Assembly

The vertical aftercooler assembly has water inlet and outlet connections on the top and bottom. The vertical aftercooler assembly is part of Cooling System Option 2.



Preparatory Steps

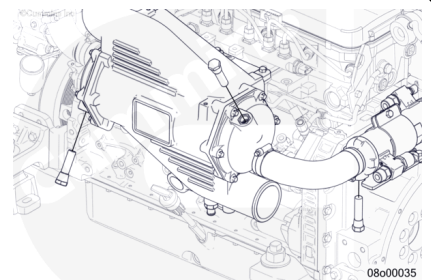
- Close the sea water supply valve(s).



Remove

⚠ CAUTION ⚠

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. If not, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the aftercooler to prevent damage to components downstream in the sea water system.



⚠ CAUTION ⚠

In some cases, it will be necessary to hold the welded fitting on the heat exchanger with an additional wrench to prevent damage to the heat exchanger.

This engine has up to five zinc plugs (one in the gear cooler, if equipped, two in the aftercooler assembly, and one or two in the heat exchanger assembly).

Shut off the sea water supply valve(s). Remove the zinc anode plugs from the bottom and top of the aftercooler assembly.

Remove the two zinc anodes from the aftercooler.

Remove the one or two zinc anode plugs from the front of the heat exchanger.

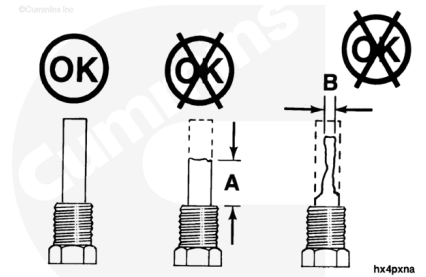
Remove the zinc anode plug from the inlet of the marine gear oil cooler, if equipped.

Inspect for Reuse

Inspect the sacrificial zinc plug. Replace as required.

Place the zinc portion of the plug on a metal surface. Strike the plug with a small hammer to remove loose material. This will help to determine the actual amount of material remaining.

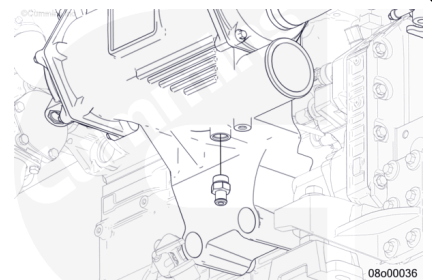
Check the length of all the zinc plugs in the heat exchanger and aftercooler, (and marine gear oil cooler, if equipped). Replace them if they are 50 percent eroded. Frequency of replacement depends upon the chemical reaction of raw water circulated through the heat exchanger and aftercooler.



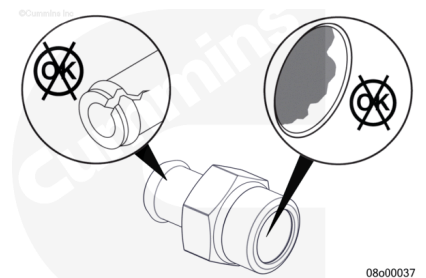
Zinc Plug Erosion Limits				
	mm		in	
A	19.0	MIN	0.750	
New		51.0	MAX	2.000
B	6.4	MIN	0.250	
New		16.0	MAX	0.630

If any zinc plug does **not** meet specification, replace it with a new zinc plug.

Remove the aftercooler check valve.



Inspect the aftercooler check valve when zinc anodes are checked. The valve should flow air freely under low pressure and close under high-pressure. A pressure of over 0.5 psi [3.45 kPa] from the threaded side of the valve should make it close. Operation can be checked using compressed air. The internal piston should rattle when shaken. Replace the valve if **not** operating properly.



Inspect the valve for excessive deposits on its interior surface. White deposits 0.06 in [1.524 mm] thick or clogging the valve indicate an internal seawater leak requiring the

aftercooler to be disassembled and inspected. Refer to Procedure 010-005 in Section 10.

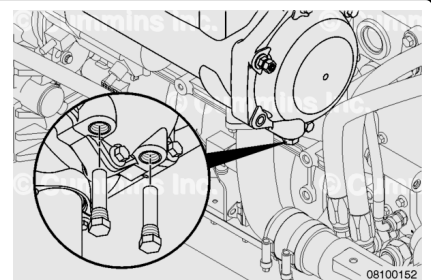
(/qs3/pubsys2/xml/en/procedures/40/40-010-005.html)

Inspect the joint between the water connections and the housing. Leaks, excessive corrosion, or salt deposits can indicate an internal seawater leak requiring the aftercooler to be disassembled and inspected. Refer to Procedure 010-005 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-005.html)

Install

⚠ CAUTION ⚠

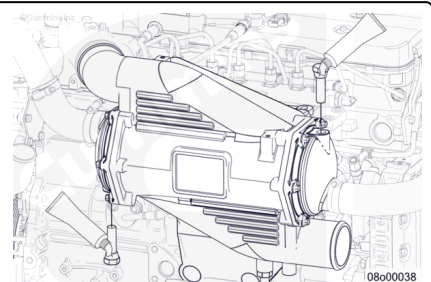
Do not use Teflon™ 2122 tape on the zinc anodes. Use will insulate the zinc anode and prevent a good ground. Use of common thread sealant that hardens will cause the anodes to break plastic end caps. Use only specified sealant.



Cooling System Option 1: Horizontal Aftercooler Assembly Zinc Anode

Apply Hercules™ Pro Dope or Rectorseal T Plus2 to the zinc anode plugs (size 1/2" x 2"). Install them in the aftercooler assembly. Hercules™ Pro Dope or Rectorseal T Plus2 **must** be used for the aftercooler assembly for Cooling System Option 1 or it will **not** seal properly.

Install the zinc anode into the aftercooler assembly.

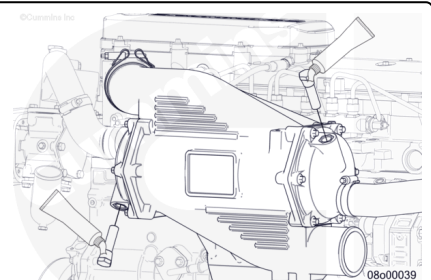


Torque Value:

Bronze Inlet and Outlet Water Connection 30 n•m [266 in-lb]

Torque Value:

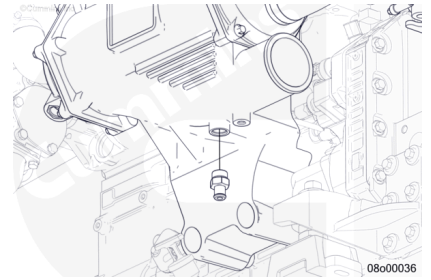
Plastic Inlet and Outlet Water Connection 9 n•m [78 in-lb]



Install a new o-ring on the aftercooler check valve, if needed.

Install the aftercooler check valve.

Torque Value: 30 n•m [266 in-lb]



Heat Exchanger Zinc Anode

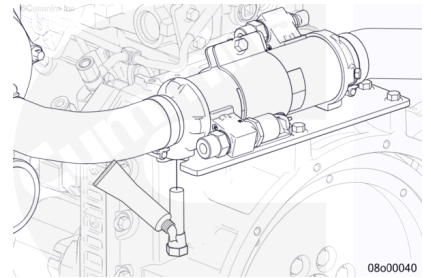
Install the zinc anode into the heat exchanger, if equipped.

Torque Value: 22 n•m [195 in-lb]

Marine Gear Oil Cooler Zinc Anode

Install the zinc anode into the marine gear oil cooler, if equipped.

Torque Value: 22 n•m [195 in-lb]



Cooling System Option 2: Vertical Aftercooler Zinc Anode

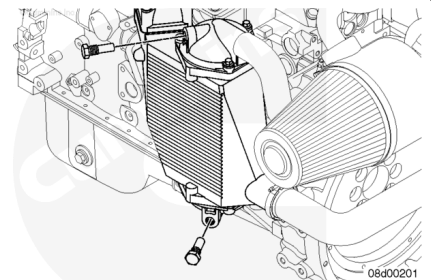
Install the zinc anode plug in the bottom of the heat exchanger.

Torque Value: 30 n•m [266 in-lb]

Install a new o-ring on the aftercooler check valve, if needed.

Install the aftercooler check valve.

Torque Value: 30 n•m [266 in-lb]



Heat Exchanger Zinc Anode

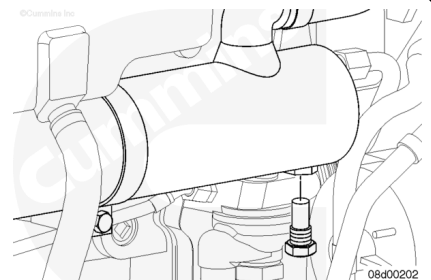
Install the zinc anode plug in the bottom of the heat exchanger.

Torque Value: 30 n•m [266 in-lb]

Marine Gear Cooler Zinc Anode

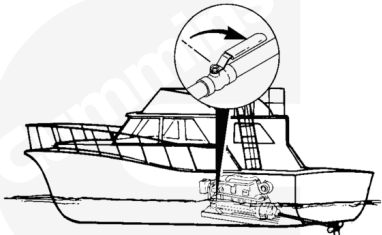
Install the zinc anode plug into the gear cooler.

Torque Value: 22 n•m [195 in-lb]



Finishing Steps

- Open the sea water supply valve(s).
- Operate the engine. Check for leaks.



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Last Modified: 09-Mar-2017
